

Rate of Water Ingress into Hollow-Core Fibre by Capillary Action

Samuel Agah*, Chiang Ping Saw, Peter Horak and Natalie V. Wheeler

Optoelectronics Research Centre, University of Southampton, SO17 1BJ, UK, *saoa1g22@soton.ac.uk

Abstract We investigate water ingress into a double-nested antiresonant nodeless fibre by capillary action, using two independent methods. We find that different parts of the fibre fill at different rates and that ingress is 2-4 times slower than into a cylindrical capillary of identical core diameter. ©2026 The Author(s)

Introduction

Recent advances in the design and fabrication of hollow-core optical fibres (HCFs) have led to major reductions in loss, and HCFs now offer the lowest attenuation of any optical fibre type [1]. These improvements are accelerating commercial adoption, with HCF cables already being deployed and carrying live traffic [2] [3]. Light is guided in an HCF by a precisely engineered microstructure formed from thin, high-purity silica membranes. In addition to the hollow core, typical HCF designs include several hollow cladding regions running continuously along the fibre length. While essential to HCF performance, these hollow regions introduce reliability considerations not present in conventional solid-core fibres.

In this work, we focus on the risk of liquid ingress, specifically water, into these hollow regions following a fibre break. Such a break could occur in various deployment scenarios: for example, if a fibre fails while submerged in surface water, or, in the future, if a transoceanic HCF cable were to break at depth. Water ingress would severely compromise transmission, as the fibre would cease to guide light; mitigation, such as removing or treating the affected fibre section, would therefore be required. To implement appropriate preventative or corrective action, it is essential to understand how factors such as fibre design and deployment environment influence the rate of liquid ingress.

We present an analytical liquid-ingress model based on flow in a cylindrical capillary and compare its predictions with experimental measurements of water ingress in a double-nested antiresonant nodeless fibre (DNANF) obtained using two independent methods. Experimentally, different sections of the DNANF are observed to fill at different rates, and both measurement approaches show that liquid ingress occurs substantially more slowly than predicted by the simple capillary model of the fibre core.

Theoretical Background and Modelling

Washburn [4] showed that the rate of ingress of liquid into a cylindrical capillary of small radius r is

given by

$$\frac{dl}{dt} = \frac{(P_c + P_d + P_g)(r^4 + 4\epsilon r^3)}{8r^2\eta l} \quad (1)$$

which depends on the total driving pressure comprising the capillary pressure P_c , the pressure difference between the outside and inside of the capillary P_d , and the pressure due to gravity P_g if the capillary is inclined at angle ϕ ; l is the length of the liquid column, η is the liquid viscosity; ϵ is the coefficient of slip at the capillary boundary (see Fig. 1).

Assuming that the capillary is laid horizontally ($P_g = 0$ Pa), and that there is a ‘no-slip condition’ ($\epsilon = 0$), integrating Eq. (1) gives

$$l(t) = \sqrt{\frac{P_c + P_d}{4\eta}} \cdot r\sqrt{t} = \sqrt{\frac{\frac{2\gamma \cos \theta}{r} + P_d}{4\eta}} \cdot r\sqrt{t} \quad (2)$$

where γ is the surface tension of the liquid, and θ is the contact angle the liquid meniscus makes with the capillary.

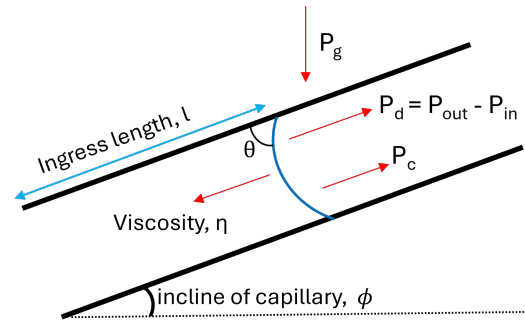


Fig. 1: Pressures that act on a column of liquid in a cylindrical capillary.

Figure 2 shows the ingress of pure water into a capillary of diameter $25 \mu\text{m}$ (with an internal pressure of 1 bar) at different external pressures. We see that at high enough external pressures, P_d becomes much larger than P_c and dominates the ingress, see Eq. (2).

Experimental Setup and Results

Two complementary experiments were carried out to measure the ingress rate of the water into a DNANF: one using a microscope and the other using an optical time-domain reflectometer (OTDR).

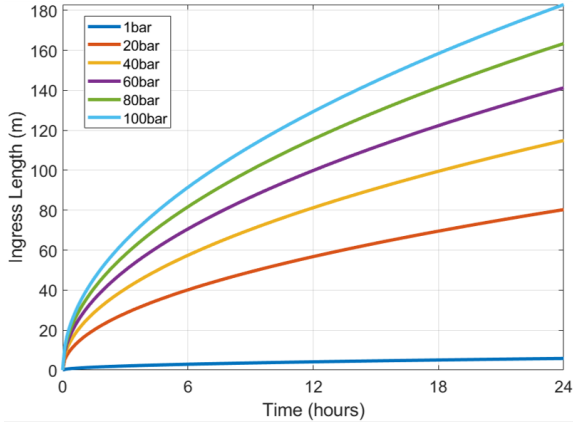


Fig. 2: Ingress of pure water ($\eta = 1.00 \text{ mPa}\cdot\text{s}$, $\gamma = 72.8 \text{ N/m}$) at varying external pressures for 24 hours into a capillary of diameter $25 \mu\text{m}$. The capillary's internal pressure is 1 bar.

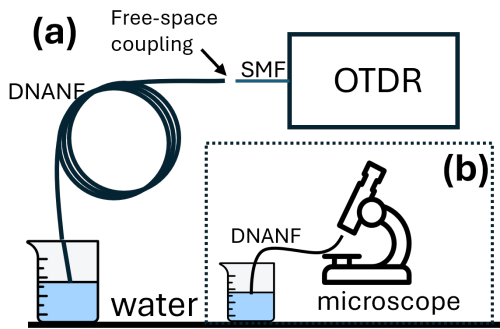


Fig. 3: (a) Setup of experiment to monitor the ingress of water into a DNANF using OTDR. (b) Setup of experiment using microscope imaging to monitor the time it takes for the water to travel through a known DNANF length.

Microscope Experiment

The first experiment used a microscope to monitor one end of a fibre (near end) of known length, while the other end of the fibre (ingress end) was submerged in water, see Fig. 3(b).

Using this method demonstrated that different sections of the DNANF fill at different rates, as shown in Fig. 4. The spaces between the outer capillaries (crevices) fill first, then the core, and eventually the cladding capillaries are filled. This is consistent with what is known about capillary action at tight corners (the crevices in this case) filling ahead of the bulk in an irregularly shaped capillary [5]. Ingress in the cladding capillaries with smaller radii is slower because the ingress length is proportional to radius according to Eq. (2).

This experiment was repeated several times

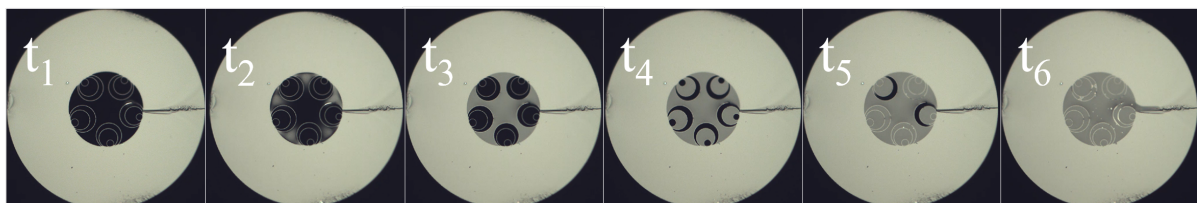


Fig. 4: Microscope images of HCF when water reaches the top face by capillary action showing that different sections of the fibre fill at different rates. The crevices are filled first at time t_2 , then the core at t_3 , before the cladding capillaries at times t_4 through t_6 .

with different fibre lengths and the time it takes for the water to travel from the ingress end to the near end was recorded to obtain a length-time plot (Fig. 5). To obtain more reliable data, time measurements of the ingress into each length were taken several times. The results show that the ingress rate into the DNANF is lower than that of the cylindrical model with diameter equal to the DNANF's core diameter. Previous work done on irregularly shaped capillaries also showed lower ingress rates [6]. We attribute this to the fact that while the larger internal surface areas of a DNANF compared to a cylindrical capillary enhance the capillary force, these large surfaces also impede laminar flow, leading overall to slower ingress rates.

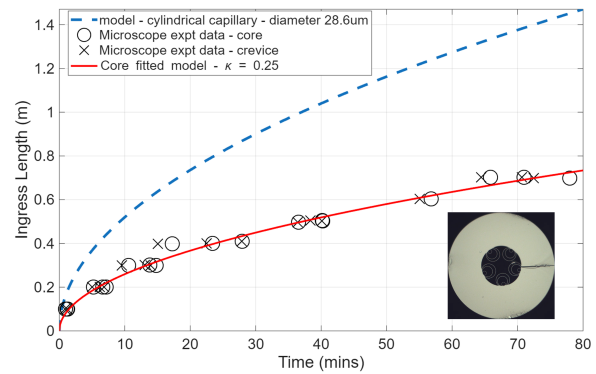


Fig. 5: Comparison of water ingress into a DNANF (into the crevice and core) with an analytic model of a capillary with the same core diameter ($28.6 \mu\text{m}$). Inset shows cross-section of DNANF.

To account for the change, a fitting factor κ is added to the ingress length equation in the absence of gravity,

$$l_{DNANF}(t) = \sqrt{\frac{\kappa \left(\frac{2\gamma \cos \theta}{r} \right)}{4\eta}} \cdot r \sqrt{t} \quad (3)$$

where r is the core radius of the DNANF.

Fitting Eq. (3) to the experimental data of Fig. 5 yields a value of $\kappa = 0.25$, meaning that the ingress length after a given time in our DNANF is half that of a capillary of the same core diameter, and filling a fibre of given length will take four times longer than a capillary. Alternatively, we could define an effective 'hydraulic radius', $r_h = \kappa r$, which is the radius a cylindrical capillary must have for the same rate of ingress as a DNANF with core radius r .

OTDR Experiment

The second experiment used an OTDR, which is normally used to obtain the loss along a fibre by measuring the backscattered light as a function of time (length). In our case, the air-water interface of the advancing water column will cause a large reflection, which will appear as a peak in the OTDR trace (see Fig. 6). It was observed that as the water advances in the fibre, the height of the reflection peak drops because, as we have seen from the microscope experiment, the water advances in the crevices faster so that the light from the OTDR initially encounters water in the crevices but not in the core (time t_2 in Fig. 4) which makes for a lossy interface, rather than a reflective one.

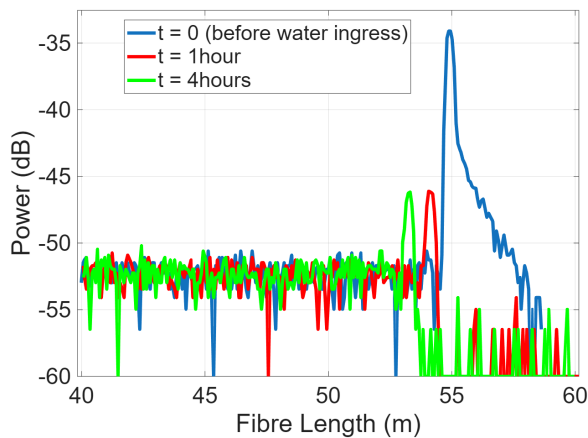


Fig. 6: OTDR trace of a DNANF before water ingress, 1 hour after ingress and 4 hours after ingress.

Taking OTDR traces as in Fig. 6 over an extended time and monitoring the position of the peaks then allows us to produce a length-time plot. The OTDR light was free-space coupled from the SMF to the DNANF to maximise the dynamic range of the OTDR measurement to ensure that the reflection peak stays above the noise floor for as long as possible. Due to the limited resolution of the OTDR, there was some uncertainty to the exact position of each peak, so a Gaussian function was fitted to each peak for a better estimate of the position of the peak along the fibre. The OTDR was set to a wavelength of 1550 nm (for most efficient guiding in the DNANF), a pulse width of 2 nm to give the best resolution available, and was averaged 4 times to get less noisy traces. A small pulse width and high average number take longer to produce a complete trace, so the OTDR was set to only measure the last 20 m of the fibre to reduce the time between traces.

The results of the OTDR experiment are shown in Fig. 7. As in the microscope experiment we observe that the water ingress is slower than predicted by Eq. (2) for a cylindrical capillary with the same core diameter. However, the fitting factor κ , Eq. (3), is 0.54. This fitting factor differs by a factor of 2 from the one calculated using the data from

the microscope experiment. The exact reasons for this discrepancy are currently unclear; contributing factors could be the slight difference in fibre design (26.6 μm and 28.6 μm core diameter, respectively), the difference in time and length scale of the two experiments, which could imply that in the microscope experiments the flow was not fully developed, or the better repeatability and better resolution of the OTDR experiment.

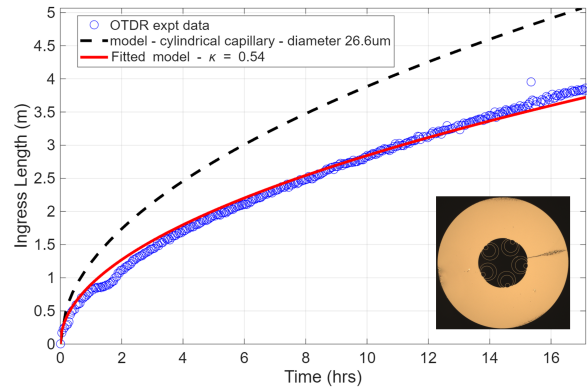


Fig. 7: Comparison of water ingress into a DNANF with an analytic model of a capillary with the same core diameter (26.6 μm). Inset shows cross-section of the DNANF.

Conclusions

We have demonstrated that water ingress into a DNANF driven solely by capillary pressure is significantly slower than ingress predicted by the simple cylindrical capillary model. Adding a fitting factor to the ingress equation, we find that for a given fibre length a DNANF takes 2-4 times longer to fill than a cylindrical capillary with a diameter equivalent to the DNANF's core diameter.

Future work will focus on understanding how the dimensions of the DNANF will affect the scaling factor in order to predict the water ingress in a given anti-resonant fibre. Work will also be done to study the flow in a DNANF driven by an external pressure, as relevant to different deployment environments.

Acknowledgements

This project was supported by the EPSRC grant FASTNET (EP/X025276/1) and an EPSRC DLA PhD studentship co-funded by Microsoft Corp.

References

- [1] M. Petrovich et al., "Broadband optical fibre with an attenuation lower than 0.1 decibel per kilometre", *Nature Photonics*, vol. 19, pp. 1203–1208, 2025. DOI: 10.1038/s41566-025-01747-5.
- [2] C. Forster. "The deployment of hollow core fiber (hcf) in azure's network". [Online]. Available: <https://techcommunity.microsoft.com/blog/azurenetworkingblog/the-deployment-of-hollow-core-fiber-hcf-in-azure%E2%80%99s-network/4395340>.
- [3] D. Ge et al., "Tb/s-class optical transmission over ar-hcfs: Field deployment, characterization, and performance", *Journal of Lightwave Technology*, vol. 44, pp. 1094–1112, 2026. DOI: 10.1109/JLT.2025.3638159.
- [4] E. W. Washburn, "The dynamics of capillary flow", *Physical Review*, vol. 17, pp. 273–283, 1921. DOI: 10.1103/PhysRev.17.273.
- [5] A. Ponomarenko, D. Quéré, and C. Clanet, "A universal law for capillary rise in corners", *Journal of Fluid Mechanics*, vol. 666, pp. 146–154, 2011. DOI: 10.1017/S0022112010005276.
- [6] J. Cai, Y. Chen, Y. Liu, S. Li, and C. Sun, "Capillary imbibition and flow of wetting liquid in irregular capillaries: A 100-year review", *Advances in Colloid and Interface Science*, vol. 304, pp. 4776–4785, 2022. DOI: 10.1016/j.cis.2022.102654.